Record Allocation for Minimizing Expected Retrieval Costs on Drum-Like Storage Devices

R.A. CODY AND E.G. COFFMAN JR.

Pennsylvania State University, University Park, Pennsylvania

ABSTRACT This paper examines the problem of distributing a set of equal-size records among the sectors of a drum-like storage device in order to exploit known access frequencies and reduce the average access time A simple catenated search model is defined for which the problem is shown to be NP-complete Heuristics are then defined and analyzed in terms of worst-case bounds. It is shown that easily implemented highest-access-frequency-first assignment rules provide an average access time very close to optimal.

KEY WORDS AND PHRASES storage allocation, record assignment on secondary storage, drum systems analysis, disk systems analysis, combinatorial allocation problems

CR CATEGORIES 4.32, 4 39, 6.35

1. *Introduction*

We consider the problem of partitioning a set of equal-length records into a set of blocks to be assigned to the sectors of a drum, or a cylinder of a disk system, so as to minimize the expected cost of latency time. Assuming a stationary probability distribution describing record access frequencies, we shall formulate the problem as a *catenated search* [1] problem Thus, we shall be considering minimum latency costs under the assumptions that a queue of requests is serviced first-in-first-out (FIFO) and that the system is operating at full capacity; i e a saturated demand for record accesses is assumed. For analyses of a number of other interesting catenated search problems the reader is referred to [1] and [6]. A particularly relevant application of our study exists in those paging systems [2] containing paging drums or disks, where the equal-length records are pages In the final section we shall be making further remarks concerning assumptions and implications of the model

We shall begin in Section 2 by defining the model and the cost function to be minimized. In this section we also verify that the basic minimization problem is NP-complete [3]. In Sections 3 and 4 we analyze a simple heuristic approach to the problem in which records are assigned to blocks largest-first, in terms of access probabilities, in the same manner described by Graham [4] for sequencing independent tasks on multiple processors with execution times known Performance bounds are derived under two assumptions: First, there is no bound on the number of records that can be assigned to a sector, and second, there exists a fixed bound k on this number for all sectors (We shall always assume that

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Authors' present addresses R A. Cody, Bell Laboratories, Holmdel, NJ 08904, E.G Coffman, Department of Computer Science, Pennsylvania State University, University Park, PA 16802

$n \leq m \cdot k$, where n is the number of records and m is the number of sectors.) We show that even in the worst case the performance of the heuristics is very close to optimal.

2. *The Model*

Let the set of n records be represented by their stationary access probabilities, $P = \{p_1, \cdots, p_n\}$. The problem of allocating these records on a drum having m sectors so that the expected latency cost is minimized consists of finding a partition, μ, of n records into m blocks, and an assignment function, π, which assigns these blocks to the m sectors on the drum. (Our terminology will be restricted to drum systems although the applications also include disk systems.) To define the cost function, we suppose an unbounded FIFO queue of requests for which the probability that the jth request is for the ith record is simply p_i for all j If S_i denotes the sum of the probabilities of records assigned to sector i, then the expected latency cost $C(\mu, \pi)$ of a record access (for all requests beyond the first) may be written as

$$C(\mu, \pi) = \sum_{i=1}^{m} \sum_{j=1}^{m} S_i S_j d_{ij}, \tag{1}$$

where d_{ij} is the latency cost incurred in the motion of the drum from the end of sector i to the beginning of sector j. Equation (1) in effect characterizes catenated search problems [1]. Assuming a unit cost for the time taken by the drum to rotate over one sector, the latency cost function will be given by

$$\begin{aligned} &\left.\begin{aligned} d_{ij} &= j - i - 1 \\ d_{ji} &= m - 2 - d_{ij} \end{aligned}\right\} \quad \text{for} \quad 1 \leq i < j \leq m, \\ &d_{ii} = m - 1 \qquad\qquad \text{for} \quad 1 \leq i \leq m. \end{aligned} \tag{2}$$

The following result simplifies our minimization problem.

LEMMA 1 *$C(\mu, \pi) = C(\mu, \pi')$ for all π and π'; that is, the assignment of sector numbers to sector blocks does not affect the expected cost.*

The details of a routine proof are omitted. It can be easily shown that for any assignment functions, π and π', that differ only by a single interchange of two sector blocks, $C(\mu, \pi) - C(\mu, \pi') = 0$. Lemma 1 then follows, since any assignment can be obtained from any other by a sequence of such simple interchanges.

In view of Lemma 1 we shall hereafter use the simpler notation $C(\mu) = C(\mu, \pi)$. Therefore, our specific problem is· Given a number, m, of sectors and a set, P, find a partition, μ, that minimizes (1) assuming the cost function (2). We may render this problem into a simpler form as follows.

Let $S = (S_1, \cdots, S_m)$, where without loss of generality, $S_1 \geq S_2 \geq \cdots \geq S_m$, and let $f(S) = \sum_{i=1}^{m} S_i^2$. Also, let $\delta_i = S_i - 1/m$, $\delta = (\delta_1, \cdots, \delta_m)$ and $f(\delta) = \sum_{i=1}^{m} \delta_i^2$.

THEOREM 1. *We have*

$$C(\mu) = (m - 2)/2 + (m/2)f(S) = (m - 1)/2 + (m/2)f(\delta), \tag{3}$$

so that the partition, μ^*, which minimizes $f(\delta)$ and $f(S)$ also minimizes the expected latency cost function, and conversely.[1]

PROOF Using the symmetry of the cost function, we may rewrite (1) as

$$C(\mu) = \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} S_i S_j (d_{ij} + d_{ji}).$$

By observing that

$$d_{ij} + d_{ji} = \begin{cases} m - 2 & (1 \leq i < j \leq m), \\ 2(m - 1) & (i = j), \end{cases}$$

[1] Note that Lemma 1 follows also from Theorem 1

we have

$$C(\mu) = (m-2)/2 \sum_i \sum_{j \neq i} S_i S_j + (m-1) \sum_{i=1}^{m} S_i^2. \tag{4}$$

Making use of

$$\sum_{i=1}^{m} S_i^2 + \sum_i \sum_{j \neq i} S_i S_j = \left(\sum_{i=1}^{m} S_i \right)^2 = 1,$$

we substitute into (4), simplify, and obtain the first part of (3). The second part of (3) follows from $f(\delta) = f(S) - 1/m$. □

The next result verifies that in fact our problem is a hard one.

THEOREM 2. *For any $m \geq 2$ the problem of constructing μ^* is NP-complete.*

The idea of the proof is to consider the problem of finding a partition for $m = 2$ such that $f(S) \leq k$, for some given k. Letting $k = \frac{1}{2}$, we see that $f(S) \leq k$ if and only if there is a partition of P into two blocks such that the block sums S_1 and S_2 are equal. It is well known that this partitioning problem is NP-complete [3]. The details of a complete proof based on this reduction are routine, and hence we leave them to the concerned reader. We should note immediately that Theorem 2 is easily extended to include the case where there is a fixed bound k on the cardinality of the blocks of the partition (see [5]).

In view of the complexity one can expect from any exact algorithm for calculating μ^*, it is reasonable to consider how well a simple heuristic might do. In the following we shall be examining two easily implemented heuristics for partitioning a set of records. Our goal is to determine how bad the heuristics can be by computing a bound to the ratio of the cost obtained by the heuristic to that of an optimal partition. For even moderately large m we shall see that the cost ratios are very close to one. Best (i.e. achievable) upper bounds are not known as yet, except for the case $m = 2$.

To motivate the study of heuristics we show that an arbitrary assignment can have a cost up to twice that of an optimal assignment. First, we need some notation and a preliminary result.

For convenience the partitioning rules to be defined will be based on the notion of assigning record access *probabilities* to blocks rather than *records* to blocks. Of course, the cost of the record distribution is completely defined by the partitioning of record probabilities. Also, for ease in reading and writing we shall refer to record access probabilities as *RP*s.

Let A be an arbitrary rule for assigning RPs to blocks, let B_i denote the set of RPs in sector (block) i, and as before let S_i denote the total probability mass in B_i $(1 \leq i \leq m)$ for the A-assignment of a given set P. Let δ_A and S_A denote the vectors obtained by applying A to P, and denote the cost ratio for two assignment rules and a given m as

$$\eta_m(A/A') = \frac{\left[\dfrac{m-1}{m} + f(\delta_A)\right]}{\left[\dfrac{m-1}{m} + f(\delta_{A'})\right]} = \frac{\left[\dfrac{m-2}{m} + f(S_A)\right]}{\left[\dfrac{m-2}{m} + f(S_{A'})\right]} \tag{5}$$

Finally, we present the following result, whose routine proof is omitted.

LEMMA 2. *Given the set $X = \{x_1, \cdots, x_m\}$ such that $\sum_{i=1}^{m} x_i = 1$ and $0 \leq x_i \leq 1$ $(1 \leq i \leq m)$, we have $1/m \leq \sum_{i=1}^{m} x_i^2 \leq 1$. Moreover, the sets X_1 such that for some i $(x_i = 1)$, $x_j = 0$ $(j \neq i)$, and X_2 such that $x_i = 1/m$ for all i achieve, respectively, the upper and lower bounds.*

THEOREM 3. *For any set P and a given m let A and A' denote two arbitrary assignment rules. Then*

$$\eta_m(A/A') \leq 2 \qquad (m \geq 2), \tag{6}$$

and this is a best bound. Moreover, (6) is a best bound if A and A' are drawn from a restricted class of assignment rules defined by a fixed block-size constraint.

PROOF. From Lemma 2 substitute upper and lower bounds on $f(S_A)$ and $f(S_{A'})$,

respectively, into (5) to obtain (6). The set for which $n = m^2$ and $p_i = 1/m$ $(1 \leq i \leq m)$, $p_i = 0$ $(m + 1 \leq i \leq m^2)$ shows that the bound is achievable in general and, in particular, under an equal block-size constraint on both A and A'. Specifically, the bound is achieved by this example when, according to A, all nonzero access probabilities are assigned to the same block, and according to A', all are assigned to distinct blocks. □

3. *Largest-First Assignment Heuristics*

We now consider an assignment rule whose execution time will have a complexity no worse than that associated with the sorting of n numbers. Basically, the rule is identical to the largest-first scheduling rule of Graham, but with the terminology changed so that "RP" is used instead of "task" and "block" is used instead of "processor." Specifically, given a set P of n RPs $p_1 \geq p_2 \geq \cdots \geq p_n$ and m $(2 \leq m \leq n)$ blocks, the rule begins by assigning the largest m RPs one to a block. The remaining RPs $p_{m+1}, \cdots, p_n$ are assigned in sequence so that p_i $(m + 1 \leq i \leq n)$ is assigned to any block in which the least probability mass has been assigned during the process of assigning $p_1, p_2, \cdots, p_{i-1}$.

The assignments resulting from the above rule will be called *L-assignments*. We shall also be interested in a modification of the rule which imposes a prespecified limit, k, on the number of RPs (and thus records) which can be assigned to a block. That is, in the final assignment we must have $|B_i| \leq k$ $(1 \leq i \leq m)$. Specifically, whenever in the above rule a block has a kth RP assigned to it, it is removed from the set of blocks to which subsequent RPs can be assigned. This will be termed the *L^*-assignment* rule and we shall assume that the given parameter k satisfies $n \leq m\ k$, so that all RPs are eventually assigned. An optimal, unconstrained assignment will be termed an *O-assignment*, and the optimal, subject to $|B_i| \leq k$ $(1 \leq i \leq m)$, will be termed an *O^*-assignment*.

Examples are shown in Figure 1 which illustrate these rules and the fact that they are not generally optimal. In this section we shall restrict ourselves to a comparison of L- and O-assignments.

In the following it will be convenient to adopt as a convention that the elements of each B_i are ordered, as in Figure 1. Thus, $B_i = (p_{i_1}, \cdots, p_{i_k})$ implies $p_{i_1} \geq p_{i_2} \geq \cdots \geq p_{i_k}$. Assuming the above ordering we introduce the notation $S(p_{i_j}) = \sum_{l=1}^{j} p_{i_l}$ and $R(p_{i_j}) = S(p_{i_j}) - p_{i_j}$. We say that p begins at $R(p)$ and terminates at $S(p)$. Also, the block-sum notation S_i will always refer to the heuristic under discussion while the primed notation S_i' refers to an optimal assignment. We shall say that an RP terminates a block if it is the last RP to be assigned to the block, see Figure 1 for example.

Our ultimate objective in this section is a bound on $\eta_m(L/O)$. For this purpose we require a number of preliminary results, most of which are simple observations which will lend insight into our combinatorial problem.

LEMMA 3. *Let p and p' be any two RPs in an L- or O-assignment and suppose p terminates a block. Then $R(p') \leq S(p)$ for all $p' \neq p$. Also, if p and p' both terminate blocks in an L^*- or O^*-assignment and if $R(p') \geq S(p)$ then p' cannot exceed any RP in the block containing p.*

Note that this result implies that $R(p) < 1/m$ for all p in L- and O-assignments. The first part of the lemma is trivial to show. A simple exchange argument suffices to prove the second part of the lemma. The following result follows easily from Lemma 3.

LEMMA 4. *In $P = \{p_1, \cdots, p_n\}$ let $P_j = \{p_1, \cdots, p_j\}$ be that subset containing just those RPs exceeding $1/m$. The L- and O-assignments both distribute the RPs in P_j one to a block, and the blocks containing these RPs have only one RP.*

We can identify the following special case in which L-assignments are optimal.

LEMMA 5. *Suppose $n \leq 2m$ for a set $P = \{p_1, \cdots, p_n\}$ and suppose the L-assignment places at most two RPs per block. Then the L-assignment must be optimal.*

PROOF. Lemma 3 can be used to show that the O-assignment must have at most two

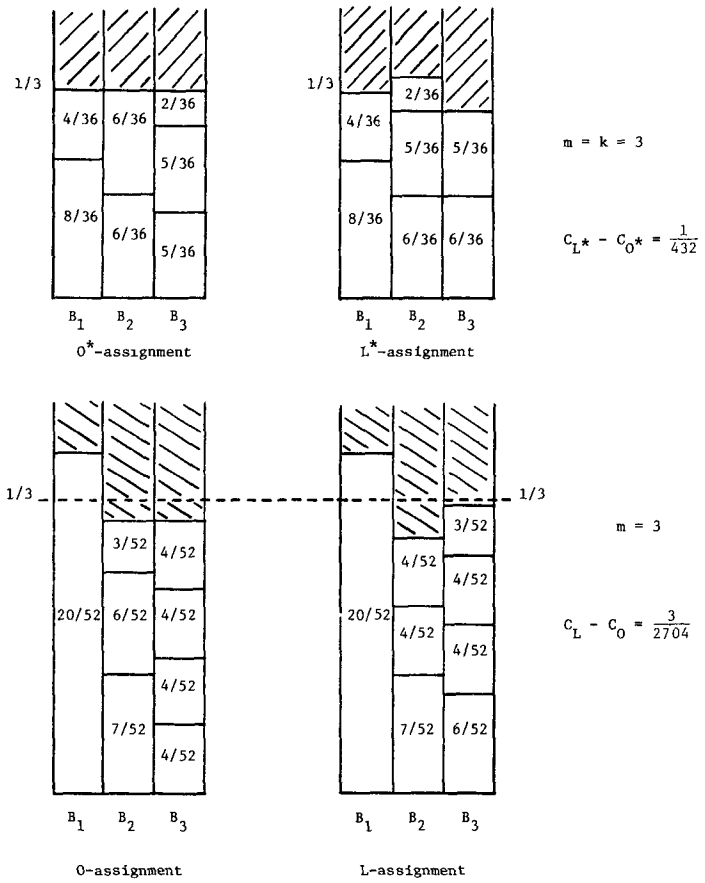


FIG 1 Example assignments

RPs per block and that the RPs in blocks having only one are all larger than those in blocks with two. A routine exchange argument then suffices to show that the blocks with two RPs in the L- and O-assignments are in one-to-one correspondence in terms of the RP pairs they contain, subject possibly to a renaming of RPs of equal size. □

As a final preliminary result we have

LEMMA 6. *Let $x_1 \geq x_2 \geq \cdots \geq x_t$ constitute a set (to be called a* profile*) of nonnegative real numbers. For a nonnegative real number b we have*

$$\sum_{i=1}^{t} (x_i - b)^2 \leq t[(a - b)^2 + \tfrac{1}{4}(x_1 - x_t)^2],$$

where

$$a = 1/t \sum_{i=1}^{t} x_i. \tag{7}$$

The proof of this result is somewhat long. For the sake of continuity we have deferred it to the Appendix.

In the main result which follows we adopt the following notion of *rank*. For a set of RPs, $p_1 \geq p_2 \geq \cdots \geq p_n$, we say that p_1 through p_m are rank-1 RPs, p_{m+1} through p_{2m} are rank-2 RPs, and so on.

THEOREM 4. *For any set $P = \{p_1, \cdots, p_n\}$ we have for $m \geq 2$,*

$$\eta_m(L/O) \leq 1 + 1/16(m - 1). \tag{8}$$

PROOF. Suppose first that in the L-assignment $v = \max_i\{S_i\} - \min_i\{S_i\} \leq 1/2m$; then by Lemma 6 we have

$$f(\delta_L) = \sum_{i=1}^{m} (S_i - 1/m)^2 \leq (m/4)(1/2m)^2 = 1/16m.$$

Thus,

$$\begin{aligned}\eta_m(L/O) &= 1 + [f(\delta_L) - f(\delta_O)]/[(m-1)/m + f(\delta_O)] \\ &\leq 1 + [m/(m-1)]f(\delta_L) \leq 1 + 1/16(m-1),\end{aligned}$$

and we are done.

Now suppose $v > 1/2m$. From the property of L-assignments in Lemma 3 we conclude that a rank-1 or rank-2 RP must terminate the L-assignment. Let P' be precisely the set of RPs in the set U of blocks whose block-sums exceed $\min\{S_i\}$ by more than $1/2m$. Assume for convenience that $U = \{B_1, \cdots, B_j\}$ $(j \geq 1)$. If U contains only blocks having exactly one RP (of rank-1 necessarily), then we may assume from Lemma 3 that $U' = \{B_1', \cdots, B_j'\}$ in the O-assignment is such that $B_i' = B_i$ $(1 \leq i \leq j)$. Moreover,

$$\sum_{i=1}^{j} (S_i - 1/m)^2 = \sum_{i=1}^{j} (S_i' - 1/m)^2,$$

and an $\eta_m(L/O) > 1$ must result from the assignments of $B_{j+1}, \cdots, B_m$.

We now want to show that $\eta_m(L/O) > 1$ must result from the assignments in $\{B_{r+1}, \cdots, B_m\}$ where $V = \{B_1, \cdots, B_r\}$ contains all those blocks having rank-2 RPs exceeding $1/2m$ or rank-1 RPs exceeding $1/m$. Since blocks with only rank-1 RPs exceeding $1/m$ can be removed from consideration by Lemma 3, we can reduce our problem immediately by assuming that each $B_i \in V$ has a rank-2 RP.

Let $P_r = \bigcup_{i=1}^{r} B_i$. Since $p > 1/2m$ for $p \in P_r$, it follows easily from Lemma 3 that each $B_i \in V$ has exactly one rank-1 RP and one rank-2 RP. Let $V' = \{B_{i_1}', \cdots, B_{i_s}'\}$ be the set of blocks in the O-assignment to which the rank-2 RPs in P_r are assigned, and let $P_s' = \bigcup_{j=1}^{s} B_{i_j}'$. We claim that $r = s$, i.e. $|V| = |V'|$. For the only way $r \neq s$ can occur is to have at least one block B' in V' contain exactly two of the rank-2 RPs in P_r. This would imply that some block B'' in the O-assignment has exactly two rank-1 RPs. Interchanging the smallest RPs in B' and B'' reduces the sum of the squares of the two block-sums and therefore contradicts the assumption of an optimal assignment. Thus, $|V| = |V'|$.

We now claim that

$$\sum_{B_i \in V} (S_i - 1/m)^2 \leq \sum_{B_i' \in V'} (S_i' - 1/m)^2. \tag{9}$$

We note from Lemma 5 that the set P' consisting of the m rank-1 RPs and just those rank-2 RPs in P_r is optimally assigned by the L-rule, in the sum-of-squares sense. The inequality in (9) follows routinely. Moreover, we can verify

$$\sum_{p \in P_r} p \leq \sum_{p \in P_r'} p. \tag{10}$$

For by definition P_r and P_r' can differ only in the rank-1 RPs they contain. But in the L-assignment P_r contains the smallest r rank-1 RPs, and therefore (10) must hold.

Without loss of generality assume that the indexing of the B_i' is such that $V' = \{B_1', \cdots, B_r'\}$ and let $S_{r+1} \geq S_{r+2} \geq \cdots \geq S_m$ and $S_{r+1}' \geq S_{r+2}' \geq \cdots \geq S_m'$. From (9) and (10) our problem has been reduced to showing that

$$\sum_{i=r+1}^{m} (S_i - 1/m)^2 - \sum_{i=r+1}^{m} (S_i' - 1/m)^2 \leq 1/16m,$$

where by our definition of r, we have $v_{r+1} = S_{r+1} - S_m \leq 1/2m$. But if we let $t = m - r$, $x_i = S_{i+r}$, $a = 1/(m-r)\sum_{i=r+1}^{m} S_i$, and $b = 1/m$ we have from Lemma 6

$$\sum_{i=r+1}^{m} (S_i - 1/m)^2 \leq (m-r)\left[1/m - [1/(m-r)] \sum_{i=r+1}^{m} S_i\right]^2 + (m-r)v_{r+1}^2/4, \quad (11)$$

where

$$[1/(m-r)] \sum_{i=r+1}^{m} S_i = [1/(m-r)]\left[1 - \sum_{p \in P_r} p\right] \leq 1/m.$$

Also, by arguments similar to those demonstrating Lemma 2, we deduce

$$\sum_{i=r+1}^{m} (S_i' - 1/m)^2 \geq (m-r)\left[1/m - [1/(m-r)] \sum_{i=r+1}^{m} S_i'\right]^2, \quad (12)$$

where

$$[1/(m-r)] \sum_{i=r+1}^{m} S_i' = [1/(m-r)]\left[1 - \sum_{p \in P_r'} p\right].$$

But from (10),

$$[1/(m-r)]\left[1 - \sum_{p \in P_r'} p\right] \leq [1/(m-r)]\left[1 - \sum_{p \in P_r} p\right] \leq 1/m,$$

and hence

$$\left[1/m - [1/(m-r)] \sum_{i=r+1}^{m} S_i\right]^2 \leq \left[1/m - [1/(m-r)] \sum_{i=r+1}^{m} S_i'\right]^2. \quad (13)$$

Finally, from (11), (12), and (13) we get

$$\begin{aligned} f(\delta_L) - f(\delta_O) &\leq \sum_{i=r+1}^{m} (S_i - 1/m)^2 - \sum_{i=r+1}^{m} (S_i' - 1/m)^2 \\ &\leq (m-r)v_{r+1}^2/4 \leq \tfrac{1}{4}m(1/2m)^2 = 1/16m. \end{aligned}$$

□

The result of Theorem 4 has not been shown to be a least upper bound on $\eta_m(L/O)$. For $m = 2$, however, we can compute this least upper bound (l.u.b.) exactly. Note that for $m = 2$, minimizing $\eta_2(L/O)$ reduces to minimizing $f(S_L)/f(S_O)$, a problem that has been studied separately, quite recently, by Chandra and Wong [6]. They have shown that maximization over y and z in Figure 2 gives the l.u.b. $[f(S_L)/f(S_O)]$. Specifically, one obtains

$$\eta_2(L/O) \leq [(y + 2z)^2 + (y + z)^2]/[(2y)^2 + (3z)^2], \quad (14)$$

where $z = (1 + 5\sqrt{13y})/27$ and $2y + 3z = 1$. Note that the l.u.b. is not achieved by a set P for which the optimal assignment is such that $f(\delta_O) = 0$. This fact along with the form of (14) are rather discouraging to anyone expecting simple l.u.b.s for arbitrary m.

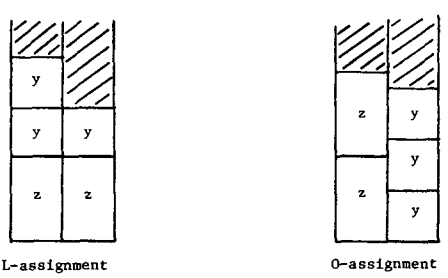


FIG. 2. Worst case for $\eta_2(L/O)$

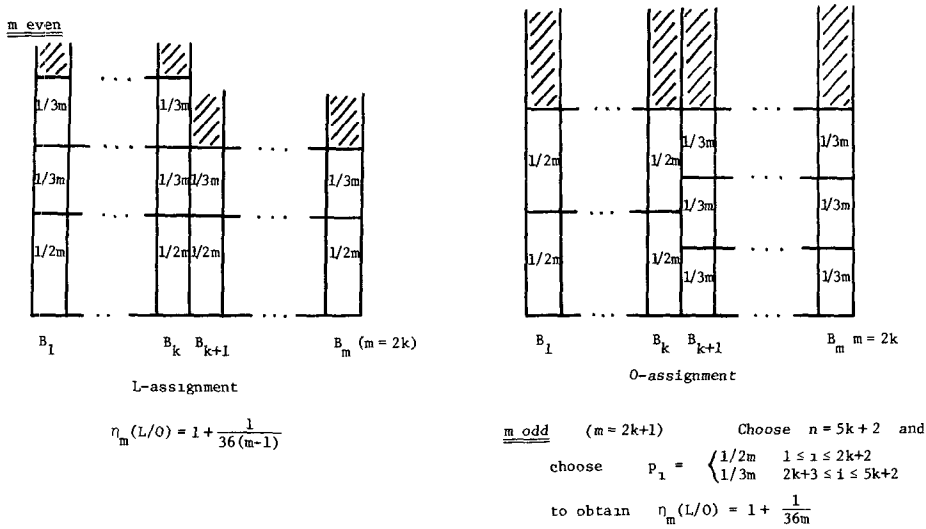


FIG 3 Example for $\eta_m(L/O)$

It is of obvious interest to determine how far (8) can be from the l.u.b. $\eta_m(L/O)$. For this purpose it is only necessary to consult the example in Figure 3 to verify that

$$1 + 1/36\,(m - 1) \le \text{l.u.b. } \eta_m(L/O) \le 1 + 1/16(m - 1). \tag{15}$$

4. *The L^*-Assignment Rule*

Clearly, for heavy use of storage we cannot assume that the bound k on the number of records that can be put on a sector has no influence. Thus, the L^*-assignment rule will be analyzed in this section, resulting in a simple bound for $\eta_m(L^*/O)$ and hence $\eta_m(L^*/O^*) \le \eta_m(L^*/O)$.

We shall assume without loss of generality that $n = k \cdot m$ (we can always add enough zero-RPs to P to ensure this without affecting the cost ratios). Before presenting the main result we need the following lemma.

LEMMA 7. *Let the (ordered) blocks $B_i = (p_{i_1}, \cdots, p_{i_k})$ $(1 \le i \le m)$, be an L^*-assignment for P $(|P| = n = mk)$. Then for any two blocks B_i and B_j we have*

$$|S(p_{i_r}) - S(p_{j_r})| \le max\,[p_{i_1}, p_{j_1}], \quad \textit{for all} \quad r = 1, 2, \cdots, k.$$

PROOF. The case $k = 1$ is trivial so suppose $k \ge 2$. We proceed by induction on r. The basis $r = 1$ is immediate, so suppose the result is true for some arbitrary r $(1 \le r < k)$, and consider $S(p_{i_{r+1}}) - S(p_{j_{r+1}})$. Without loss of generality let $S(p_{i_{r+1}}) > S(p_{j_{r+1}})$ and suppose contrary to the lemma that $S(p_{i_{r+1}}) > S(p_{j_{r+1}}) + \max\,[p_{i_1}, p_{j_1}]$, and hence $R(p_{i_{r+1}}) > S(p_{j_{r+1}})$. Since the B_is constitute an L^*-assignment we must have $p_{i_{r+1}} \le p_{j_{r+1}}$. Therefore, $S(p_{i_{r+1}}) - p_{i_{r+1}} > S(p_{j_{r+1}}) - p_{j_{r+1}} + \max\,[p_{i_1}, p_{j_1}]$ and $S(p_{i_r}) > S(p_{j_r}) + \max\,[p_{i_1}, p_{j_1}]$, which contradicts the inductive hypothesis. □

THEOREM 5. *For all sets P and parameters m, k, and $n = m \cdot k$ we have*

$$\eta_m(L^*/O^*) \le \eta_m(L^*/O) \le (m + 1)/(m - 1). \tag{16}$$

PROOF. From the definition of $\eta_m(L^*/O)$ provided by (5) we can write

$$\begin{aligned}\eta_m(L^*/O) &\le 1 + [f(\delta_{L^*}) - f(\delta_O)] \Big/ \left[\frac{m-1}{m} + f(\delta_O)\right] \\ &= 1 + \left[\sum_{i=1}^{m} S_i^2 - \sum_{i=1}^{m} (S_i')^2\right] \Big/ \left[\frac{m-1}{m} + f(\delta_O)\right].\end{aligned}$$

From the last expression we justify

$$\eta_m(L^*/O) \leq 1 + [m/(m-1)]\sum_{i=1}^{m} (S_i - S_i')(S_i + S_i'). \tag{17}$$

If we can show that $S_i - S_i' \leq 1/m$ $(1 \leq i \leq m)$, then from (17) we can write

$$\eta_m(L^*/O) \leq 1 + [m/(m-1)](1/m)\sum_{i=1}^{m} (S_i + S_i') = 1 + 2/(m-1),$$

from which (16) follows directly. Thus, we shall be done once we show that $S_i - S_i' \leq 1/m$.

As in Theorem 4 let $p_1 \geq p_2 \geq \cdots \geq p_j$ be just those RPs greater than or equal to $1/m$, and suppose they are assigned respectively to $B_1, \cdots, B_j$ in the L^*-assignment and to $B_1', \cdots, B_j'$ in the O-assignment. Also, assume $S_{j+1} \geq S_{j+2} \geq \cdots \geq S_m$ and $S_{j+1}' \geq S_{j+2}' \geq \cdots \geq S_m'$. In the L^*-assignment no RP assigned beyond $1/m$ can be larger than any RP which terminates before $1/m$. It follows easily that $S_i - S_i' \leq S_m$ $(1 \leq i \leq j)$, and since $S_m \leq 1/m$, we have $S_i - S_i' \leq 1/m$ $(1 \leq i \leq j)$. It remains, therefore, to consider $B_{j+1}, \cdots, B_m$ and $B_{j+1}', \cdots, B_m'$.

We can verify, first of all, that $S_{j+1} - S_m \leq 1/m$ from Lemma 7. Now let r be the smallest integer such that $S_r - S_r' > 1/m$. (Clearly, $S_r > 1/m$.) Since $S_r - S_m < 1/m$ and since $S_i' \leq S_r'$ $(i > r)$, it must be that $S_i' < S_m$ $(i \geq r)$. But $S_i' < S_m$ $(i \geq r)$, and $S_r > 1/m$ imply the existence of some $l(j < l < r)$ such that $S_l' \geq S_r > 1/m$, in order that the total probability mass be 1 in the O-assignment. (Clearly, $l \leq j$ is impossible since in any O-assignment $R(p) < 1/m$ for all p.) Hence, $S_l' - S_m' > S_r - S_m' > 1/m$. This in turn implies the existence of an RP larger than $1/m$ but assigned in $B_{j+1}', \cdots, B_m'$, or an RP, p, such that $R(p) > S_m'$. The first event contradicts our definition of $B_1', \cdots, B_j'$ and the second violates Lemma 3. Hence, $S_i - S_i' < 1/m$ $(1 \leq i \leq m)$. □

Once again it is possible to obtain an l.u.b. for the case $m = 2$. Omitting the details, it can be shown that one need simply maximize $\eta_2(L^*/O)$ over t in Figure 4 and then take the limit $p \to 0$ and $k \to \infty$ to obtain

THEOREM 6. *For $m = 2$ and an arbitrary set P we have*

$$\eta_2(L^*/O) \leq \tfrac{1}{4}(3 + \sqrt{5}),$$

and this bound is best possible.

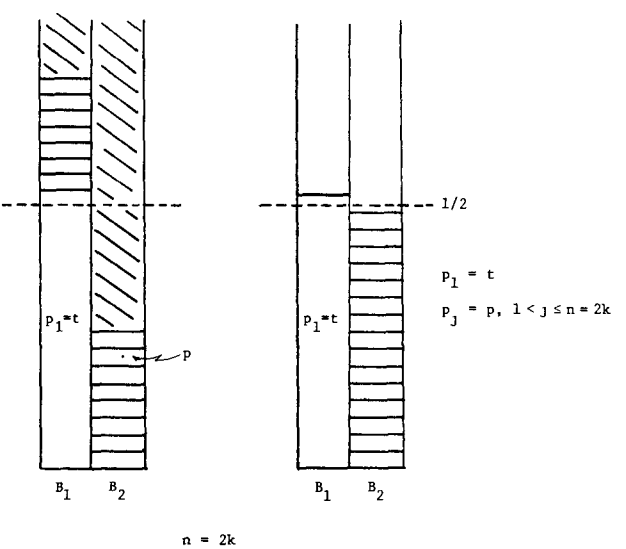
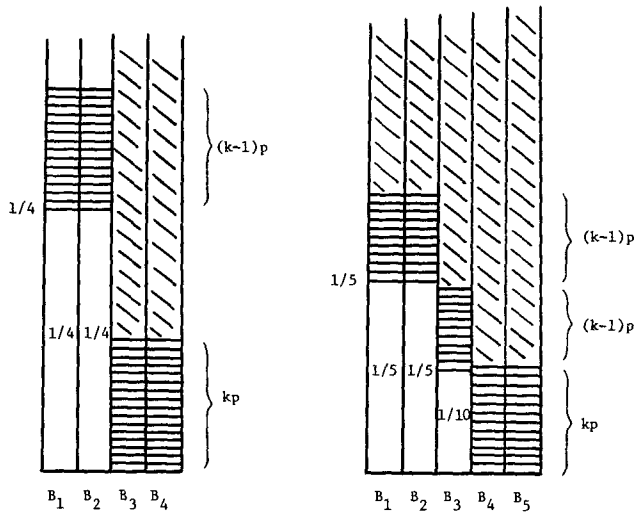


FIG. 4. Worst case for $\eta_2(L^*/O)$



p chosen so that $\sum_{1}^{n} p_i = 1$

FIG. 5. Example for $\eta_m(L^*/O)$

By generalizing the examples shown in Figure 5 for $m = 4$ and $m = 5$ it is not difficult to verify for arbitrary m

$$1 + 1/4(m - 1) \leq \text{l.u.b. } \eta_m(L^*/O) \leq (m + 1)/(m - 1).$$

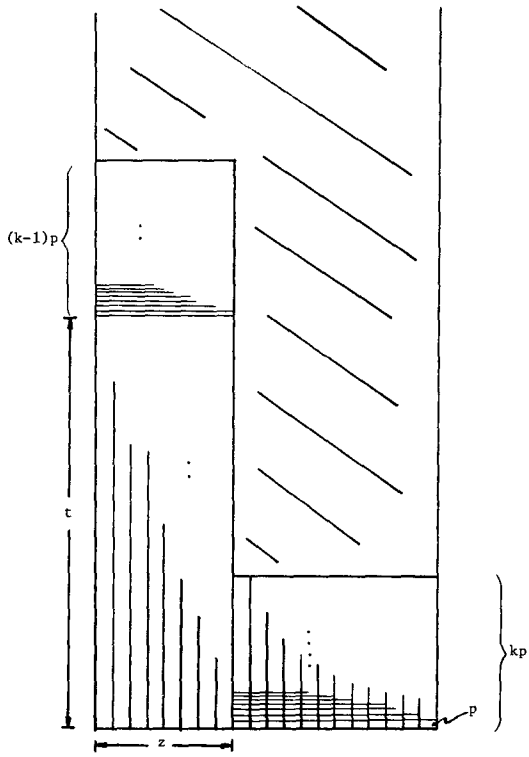
5 *Final Remarks*

Fixing a scheduling rule then optimizing the allocation of records, and fixing the allocation then optimizing the scheduling rule, are two complementary approaches to improving the performance of secondary storage systems. A typical example of the latter approach has appeared in the analyses of shortest-access-time-first sequencing policies for devices with rotational latency [2]. In this paper we have chosen the former approach to the same type of devices and shown that very simple allocation heuristics yield average access times very close to minimal.

In both approaches stationary access frequencies have been assumed known, thus implying, for example, fixed or archival-type data bases or subroutine libraries for which a priori or performance-measurement statistics are available. Of course, slowly varying access frequencies can be accommodated in a system which performs dynamic reallocation based on accumulated data on access frequencies. Moreover, performance can be improved significantly even with somewhat inaccurate estimates of access frequencies although quantitative measures of such improvements would likely be rather difficult to obtain within a reasonable analytical model

It is also the case that precise results in both approaches normally are based on the assumption of equal record sizes. The applicability to paging systems is immediate In our model the assumption can be relaxed with little or no change in the analysis only if variable length records are stored in their entirety in the same (sector) block, a property more likely to apply to disk systems than drum systems.

In our model the assumption of a system operating at capacity needs little justification, for it is this property we wish to maximize in those frequently encountered systems where auxiliary storage constitutes a bottleneck in overall system operations. We have also assumed as our fixed scheduling rule the common FIFO rule This assumption suggests a



t,z,p chosen so that total probability mass is 1

FIG 6. Example for l u b $\eta_m(L^*/O)$

source for further research, particularly into those systems where we are free to decide on both the allocation scheme and the scheduling rule.

From an engineering point of view, we have established that L- and L^*-assignments provide efficient partitioning algorithms for achieving nearly minimal access times, especially for moderately large m. From a mathematical point of view, however, open problems remain. Certainly, one is led to expect that both bounds in Theorems 4 and 5 can be tightened. The bound in Theorem 5 is the cruder of the two, and because of the inherent properties of the problem, it appears the simpler one to work on. In a future paper we hope to show that examples of the sort shown in Figure 6, where $\eta_m(L^*/O)$ is maximized over t and z, provide bounds for $\eta(L^*/L)$ and $\eta_m(L^*/O)$ which are extremely close to the best bounds.

Appendix. Proof of Lemma 6

First, from $x_i - b = (x_i - a) + (a - b)$ we can write

$$\sum_{i=1}^{t} (x_i - b)^2 = t(a - b)^2 + 2(a - b)\sum_{i=1}^{t} (x_i - a) + \sum_{i=1}^{t} (x_i - a)^2.$$

The second term vanishes because of (7), so that we need only show that

$$\sum_{i=j+1}^{t} (x_i - a)^2 \leq (t/4)(x_1 - x_t)^2. \tag{A1}$$

Let $v = x_1 - x_t$, $x_1 - a = \alpha v$, and $a - x_t = (1 - \alpha)v$ $(0 < \alpha < 1)$. Let j be defined such that $x_j > a$ and $x_{j+1} \leq a$. We proceed by considering two symmetric cases.

Case 1. $\alpha j > (1 - \alpha)(t - j)$. In this case we can write

$$\sum_{i=1}^{t} (x_i - a)^2 \leq j\alpha^2 v^2 + \sum_{i=j+1}^{t} (x_i' - a)^2, \tag{A2}$$

where $x_i' \leq x_i$ $(j + 1 \leq i \leq t)$, and $\sum_{i=j+1}^{t} (a - x_i') = j\alpha v$. We know such a collection $\{x_i'\}_{i=j+1}^{t}$ exists because the assumption of case 1 implies $j\alpha v < (t - j)(1 - \alpha)v$. Figure 7 shows that profile transformation corresponding to (A2).

Next, suppose for some $i < l$ that $x_i' > x_l' > x_t$. Now define $x_l'' = x_l' - \min\{x_l' - x_t, a - x_i'\}$ and $x_i'' = x_i' + [x_l' - x_l'']$, thus retaining

$$\sum_{i=j+1}^{t} (a - x_i'') = \sum_{i=j+1}^{t} (a - x_i').$$

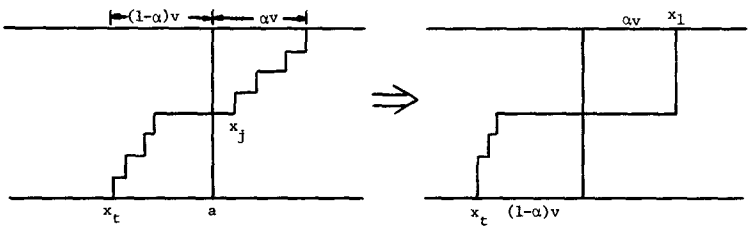


FIG 7 Example transformation

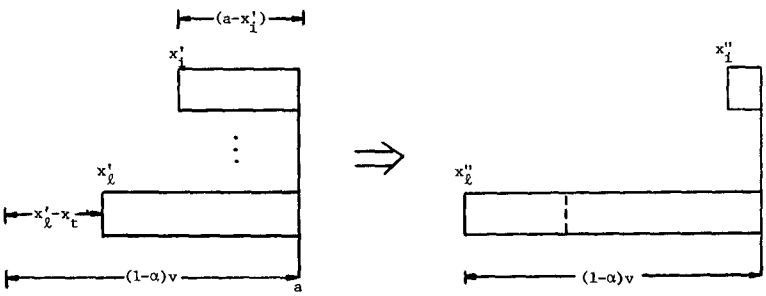


FIG 8 A second transformation

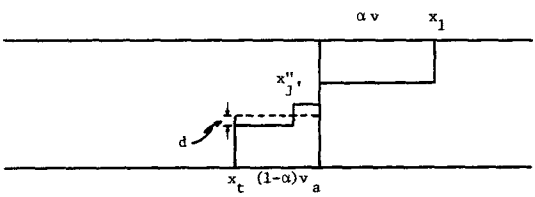


FIG 9 A third transformation

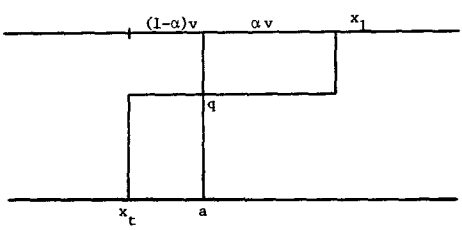


FIG. 10 A fourth transformation

It follows simply from the relations $x_l'' = x_l' - c$, $x_i'' = x_i' + c$, for some $c > 0$ and $x_l' < x_i'$ that

$$(a - x_l'')^2 + (a - x_i'')^2 \geq (a - x_l')^2 + (a - x_i')^2.$$

(This type of transformation is pictured in Figure 8.) If we apply this transformation exhaustively in the order $x_{t-1}', x_{t-2}', \cdots, x_{j+2}'$ we shall obtain a j' $(j + 1 \leq j' \leq t)$ such that $x_t \leq x_{j'}'' \leq a$, and $x_{j'+1}'' = \cdots = x_t'' = x_t$, as shown in Figure 9. To continue producing a worse case we regard the index set $\{1, 2, \cdots, t\}$ as the real interval $(0, t)$ and the profile a function $x(q)$ of the continuous variable q $(0 \leq q \leq t)$. This motivates our replacement of the $(a - x_{j'}'') \times 1$ rectangle by the rectangle $(1 - \alpha)v \times d$, where $d = (a - x_{j'}'')/(1 - \alpha)v$ (see Figure 9). Since $d(1 - \alpha)^2v^2 > (a - x_{j'}'')^2$ we can write

$$\sum_{i=j'}^{t} (a - x_i')^2 \leq [t - (j' - d)](1 - \alpha)^2v^2,$$

where $(t - j' + d)(1 - \alpha)v = j\alpha v$.

As a last step we find that point q which divides the interval $(j, j' - d)$ in the proportion $(1 - \alpha)/\alpha$ and extend the rectangles up to q as shown in Figure 10. Finally, we can write

$$\sum_{i=1}^{t} (x_i - a)^2 \leq q\alpha^2v^2 + (t - q)(1 - \alpha)^2v^2, \tag{A3}$$

where $q\alpha v = (t - q)(1 - \alpha)v$ or

$$q = t(1 - \alpha). \tag{A4}$$

Case 2. This case is treated in a fashion that is exactly symmetric to case 1. Again we arrive at (A3).

Using (A4) in (A3) we get

$$\sum_{i=1}^{t} (x_i - a)^2 \leq t\alpha^2(1 - \alpha)v^2 + t\alpha(1 - \alpha)^2v^2 = t\alpha(1 - \alpha)v^2.$$

This function of α is maximized at $\alpha = \frac{1}{2}$ and hence (A1) follows immediately. □

It may be observed that the bound in (A1) is achievable for t even, but not when t is odd.

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